

Akshat Gupta — Engineering Dashboard (All-in-One)

 **Akshat Gupta** — Software Engineer · Backend · Full-Stack · AI & Observability | Full-time at **Mark AI** · B.Tech CSE (AI), KIET (AKTU) | GitHub @akshatgg (github.com/akshatgg)

An all-in-one dashboard of everything I have built, the stack I work in, and the experience I have gained — on one page.

Key Metrics

Metric	Value	Where
 Pull requests authored	102 (100 merged)	Mark AI — current platform
 Commits, sole engineer	345 across 9 repos	Mark AI (Adneuron org)
 Pull requests authored	244 (215 merged)	VibeMonitor
 Microservices under unified observability	15+	VibeMonitor
 Cloud / observability integrations built	10+	VibeMonitor
 Personal public repos	77 (since 2022)	GitHub @akshatgg
 Roles held	7	FTE x2 · Co-founder · Freelance · Intern → Mentor

Tech Stack — at a glance

-  **Languages** — Python · TypeScript · JavaScript · Kotlin · Java · SQL · Go · Dart · C/C++
-  **Backend & APIs** — FastAPI · SQLAlchemy 2.0 async · Alembic · Pydantic · Node/Express · OAuth2 / JWT · WebSocket / SSE · Stripe · Razorpay
-  **Mobile / Digital Signage** — Kotlin Android · ExoPlayer · WorkManager · PixelCopy · AOSP (no Play Services) · Proof-of-play
-  **Frontend** — Next.js 15 · React 19 · Tailwind 4 · Radix / shadcn · Zustand · TanStack Query · Three.js / WebGL · TipTap · Recharts · Angular
-  **AI / LLM** — LangGraph · LangChain · Tool-calling agents · LangFuse · OpenAI · Gemini · Azure OpenAI · RAG · Playwright NL→script · tree-sitter · Hunyuan3D · LaMa
-  **Databases & Storage** — PostgreSQL · Cloud SQL · ClickHouse · MongoDB · Redis · Firebase · Prisma · AWS S3 · GCS + signed URLs
-  **Observability** — Prometheus · Grafana · Datadog · New Relic · CloudWatch · OpenTelemetry · Sentry · RED & USE
-  **Cloud & DevOps** — AWS (EKS · ECS · RDS · IAM · S3) · GCP (Cloud Run · Cloud SQL · GCS) · Cloud Run GPU (L4) · Workload Identity Federation · Azure · Docker · Kubernetes · GitHub Actions CI/CD

🔒 **Security & Compliance** — SOC 2 · Dependabot · Bandit / pip-audit · Gitleaks · PII redaction (Presidio) · SSRF guard · Turnstile · Signed device payloads

🔗 **Integrations built** — GitHub App/OAuth · Slack · Grafana · Datadog · New Relic · CloudWatch · Azure · GCP · K8s · Stripe · Razorpay · Twilio · Gemini API · MindAR · PostHog

🟢 What I Build at Mark AI (CURRENT)

Software Engineer, full-time, Jun 2026 → · AI-native DOOH advertising platform · sole engineer on the whole stack

Advertisers book ad slots that play on physical Android signage screens Mark AI owns and operates. I build all of it: the backend, the player on the screens, the operator CMS, the advertiser platform and the AI on top.

- **1 · Signage backend** — FastAPI + PostgreSQL, 20+ domain modules, 40 Alembic migrations. Removed the WebSocket in favour of a stateless heartbeat contract (`register/heartbeat/schedule/proof`) so the fleet scales horizontally.
- **2 · Kotlin Android player** — runs fullscreen 24/7 on AOSP boxes with no Play Services; zero-touch enrolment from a key baked into a per-owner build, re-registers when its token expires, operating-hours gating, and location even on an Ethernet-only screen.
- **3 · Proof-of-play** — the billing trail. Plays attributed by booking id, milestone screenshots armed on the playable timeline, captures on the render callback (never a timer, never half-rendered, never twice), GPS overlay burned into each frame.
- **4 · AI booking co-pilot** ★ — books campaigns from plain English: tool-calling over the live domain, two guardrail layers, a deterministic slot-capture backstop so it never re-asks, per-screenowner scoping, proof-of-play gap analysis. Tested with an actor/verifier agent harness.
- **5 · Location & ranking engine** — typo-tolerant word-coverage scorer (ported to the frontend, guarded by a shared golden table), haversine distance, forward geocoding, fleet bbox, misspelling correction.
- **6 · Operator CMS** — bookings, fleet monitoring, playback loops, operating hours, content, API keys, reports; superadmin / screenowner roles with full parity.
- **7 · Advertiser platform** — Next.js 15 marketing site with a 3D city hero, email-first auth, booking wizard, a live slot map, proof-of-play reports and a screenowner console, on its own FastAPI service that plugs into the fleet via the partner API.
- **8 · Booking approvals & payments** — a four-part approval workflow (bookings wait for the screen owner), Razorpay, campaign PDF reports and full proof-of-play Excel exports.
- **9 · Partner API** — a published `/api/ext/*` API-key surface with per-key attribution, external advertiser refs, slot occupancy and partner pause/resume/cancel.
- **10 · Creative Fit** — refits a finished ad to any screen resolution: Gemini layout blueprint + LaMa inpainting + Pillow relayout, output kept native/4K.
- **11 · 3D & AR** — self-hosted Hunyuan3D on a Cloud Run L4 GPU, plus an AR showroom for previewing a campaign in a real space.
- **12 · Cloud & delivery** — Cloud Run + Cloud SQL, Workload Identity Federation (no SA keys), signed URLs (no public buckets), direct-to-GCS uploads, and a secrets repo whose CI applies service config on push to `main`.

📖 Full breakdown with representative PRs: **Mark AI — What I Built (Current Role)**

What I Built at VibeMonitor

Software Engineer, full-time, Sep 2025 – Jun 2026 · AI-driven observability platform · 244 PRs

- **1 · CI/CD & DevOps** — multi-stage GitHub Actions, ~20% faster builds, ECS deploys, Alembic migration gate, RDS snapshot automation.
- **2 · Integrations (GitHub · Slack · Grafana)** — GitHub App, tools router, webhooks, OAuth2+PKCE; Slack install/uninstall; Grafana.
- **3 · Observability providers** — credential-verified CloudWatch, New Relic, Datadog, Azure Monitor, GCP & Kubernetes — logs + metrics, encrypted creds, health checks.
- **4 · Agentic RCA engine 🤖** — LangGraph agent tracing incidents through telemetry + source code (tree-sitter); HolmesGPT benchmark at ~90%.
- **5 · Authentication & security** — Google & GitHub OAuth (PKCE), credential auth, email verify/reset, HttpOnly refresh tokens, Remember-Me, Google One-Tap.
- **6 · Billing (Stripe)** — subscriptions, plan up/downgrade, service & per-key rate limits, weekly AI-usage limits.
- **7 · Teams · Superadmin · Enterprise** — teams schema + CRUD module, enterprise-workspace management, superadmin tooling.
- **8 · Blog platform** — TipTap rich editor with autosave, tables, code themes, S3 image proxy, preview & author roster.
- **9 · Log-ingestion platform** — high-throughput ingestion into ClickHouse, LogExplorer with bounded queries; authored the Python & JS SDKs.
- **10 · Beryl — AI test automation ⭐** — crawl→NL→Playwright→run pipeline with Redis pub/sub, worker-pool parallelism, auto-retry, per-step screenshots, auto-login crawl, SSRF guard & captcha.
- **11 · Frontend platform & UX** — large parts of the Next.js app; led the Redux→Zustand migration; dark mode, responsive landing pages, Error Boundaries.
- **12 · Email infrastructure** — Mailgun with scheduled sends & templates; migrated transactional email to Postmark.

Professional Experience

Role	Org	Type	Period	Headline
Software Engineer	Mark AI	FULL-TIME · CURRENT	Jun 2026 –	Sole engineer on an AI-native DOOH platform — FastAPI/Postgres backend, Kotlin player APK, operator CMS, advertiser platform, AI co-pilot; 102 PRs.
Software Engineer	VibeMonitor	FULL-TIME	Sep 2025 – Jun 2026	244 PRs across backend + frontend; RCA agent, integrations, Stripe, Beryl.
Full Stack Developer	Sakhi Women	FREELANCE	Jan – Feb 2026	Ride-booking; Next/Express/Mongo; Twilio WhatsApp; ~70% lower query latency.
Full Stack Developer	Mark AI	FREELANCE	Dec 2025 – Jan 2026	Earlier DOOH product; Next/FastAPI/Mongo; Xibo-based CMS integration; round-robin load balancing.

Role	Org	Type	Period	Headline
Full Stack & Server Dev	iTax Easy	INTERNSHIP	Feb – Sep 2025	GST/ITR tax platform; PostgreSQL/Next.js/Prisma; ~45% faster loads, hPanel hosting.
Co-Founder & Tech Lead	PrepSaarthi	STARTUP	Nov 2024 – Jul 2025	Edtech; React+MongoDB; ₹1 Lakh revenue; Vite migration (+40% engagement).
Frontend Engineer → Mentor	Al-Zira Technology	INTERNSHIP	Feb – Dec 2024	React client projects & UI animations; promoted to Mentor.

📌 **iTax, Al-Zira, Sakhi & the earlier Mark AI engagement are company / freelance work** — their repos sit on my GitHub but they are *not* personal projects, so they live here under Experience.

🌟 Personal Projects

🚀 Cosmic-Trek (Orrey App) · FLAGSHIP

Three.js · WebGL · GSAP · Vite · RAG. 3D solar-system simulator (8+ bodies, 50+ exoplanets), satellite-launch physics, and a RAG chatbot grounded in the Orbital-Mechanics textbook. Repo: github.com/akshatgg/OrreyApp

✈️ UpInTheSky

React · Firebase · Firestore · Tailwind. Full travel-booking frontend with real-time Firestore data and Firebase Auth; 100+ listings.

⚖️ Legal E-Library

Angular/Next.js · Node/Express · Prisma · PostgreSQL. Searchable case-law library; 10,000+ docs across 50+ categories. Repo: github.com/akshatgg/E_Library

🛒 SipCraft

Vanilla HTML/CSS/JS · Stripe. Full e-commerce site from scratch — cart, live totals, checkout via pure DOM — with end-to-end Stripe payments. Repo: github.com/akshatgg/SipCrafts

🔗 Deep-dive pages

This dashboard summarises everything; the detailed pages have the full breakdown:

- 🟢 Mark AI — What I Built (current role, with representative PRs)
- 🧠 Knowledge Gained — full point-wise inventory
- 🚀 VibeMonitor — What I Built (with representative PRs)
- 👤 Professional Experience — full bullets
- 🌟 Personal Projects

Compiled 4 June 2026 · updated 7 August 2026 from the Adneuron org (Mark AI: 9 repos, 345 commits, 102 PRs), GitHub (akshatgg + akshatgggg / Vibe-Monitor org), résumé and 1:1 work records.